

```
rep(5,3)
```

```
[1] 5 5 5
```

Rules for object names in R

1. Object names are case sensitive: object MyObject is not the same as myobject
2. They can contain both numbers or letters but should START with a letter, e.g. 2my-object is not a valid object name.
3. You can use underscores and points. Other special characters should be avoided.
4. Avoid using function names as object names.

Vector Operations

You can add, subtract, multiply and divide vectors.

```
vector1 <- c(1,2,3,4,5)
vector2 <- c(2,3,4,5,6)
# addition
vector1 + vector2
```

```
[1] 3 5 7 9 11
```

```
# subtraction
vector1 - vector2
```

```
[1] -1 -1 -1 -1 -1
```

```
# multiplication
vector1*vector2
```

```
[1] 2 6 12 20 30
```

```
# division
vector1/vector2
```

```
[1] 0.5000000 0.6666667 0.7500000 0.8000000 0.8333333
```

Exercise Create the following vectors in R using `seq()`

- i. 1, 1.5, 2, 2.5, ..., 12
- ii. 1, 8, 27, 64, ..., 1000

Subsetting Vectors

It's frequently necessary to extract some of the elements of a larger vector.

In R you can use square brackets to select an individual element or group of elements:

We create a vector of ages and see how we can subset the vector

```
age <- c(18, 23, 24, 26, 27, 29)

age
```

```
[1] 18 23 24 26 27 29
```

We can extract the third element of the vector `age`

```
# extract third element
age[3]
```

```
[1] 24
```

Extracting the first three elements of the vector

```
age[1:3]
```

```
[1] 18 23 24
```

We can exclude an element from a vector as follows:

```
age[-2] # excluding the second element
```

```
[1] 18 24 26 27 29
```

Exercise The built-in vector `LETTERS` contains the uppercase letters of the alphabet. Produce a vector of

- i. first 12 letters
- ii. odd numbered letters
- iii. English consonants

Numerical Summaries of Vectors

Measures of Central Tendency

Mean The mean (or average) is the most popular and well known measure of central tendency. It can be used with both discrete and continuous data, although its use is most often with continuous data

$$\bar{x} = \frac{x_1 + x_2 + \dots + x_n}{n}$$

where n denotes the number of observations in the vector

Example

Consider the data in the table below

65	55	89	56	35	14	56	55	87	45	92
----	----	----	----	----	----	----	----	----	----	----

Calculate the mean both by hand and in R

```
# input the values

y <- c(65,55,89,56,35,14,56,55,87,45,92)

# mean

mean(y)
```

```
[1] 59
```

```
# by hand

sum(y)/length(y)
```

```
[1] 59
```

Harmonic Mean Harmonic mean is defined as the average of the reciprocal values of the given values. That is for finding the harmonic mean of the given data set we first take the reciprocal of the data set and then divide the number of the data set by the sum of the reciprocal values.

The formula is

$$HM = \frac{n}{\frac{1}{x_1} + \frac{1}{x_2} + \dots + \frac{1}{x_n}}$$

```
harmonic_mean <- length(y)/sum(1/y)
harmonic_mean
```

```
[1] 45.21455
```

Geometric Mean The Geometric Mean (GM) is the average value or mean which signifies the central tendency of the set of numbers by taking the root of the product of their values.

$$GM = (x_1 \times x_2 \times \dots \times x_n)^{\frac{1}{n}}$$

```
geometric_mean <- prod(y)^(1/length(y))
geometric_mean
```

```
[1] 53.21217
```

Median The median is the middle score for a set of data that has been arranged in order of magnitude.

If n is even then the median is given by the mean of $(\frac{n}{2})^{th}$ observation and the $(\frac{n}{2} + 1)^{th}$

```
median(y)
```

```
[1] 56
```

Mode The mode is the most frequent score in our data set. The simplest way is to get the table of values.

```
w <- c(1,1,2,3,4,5,6)
table(w)
```

```
w
1 2 3 4 5 6
2 1 1 1 1 1
```

Quartiles Quartiles are the values from the dataset which divide the dataset into four equal parts where each part of the dataset contains an equal number of observations.

When n is odd we have:

The first quartile Q_1 is given by

$$Q_1 = (\frac{n+1}{4})^{th} \text{ term}$$

Second Quartile

$$Q_2 = (\frac{n+1}{2})^{th} \text{ term}$$

Third Quartile

$$Q_3 = (\frac{3(n+1)}{4})^{th} \text{ term}$$

To get the quartiles in R use the `summary()` function

```
vec <- c(19, 20, 21, 22, 23, 24, 25, 26, 27)
summary(vec)
```

Min.	1st Qu.	Median	Mean	3rd Qu.	Max.
19	21	23	23	25	27

When n is even we have:

$Q_1 = \text{the value of } \frac{n}{4} \text{ observation}$

$Q_2 = \text{the value of } \frac{n}{2} \text{ observation}$

$Q_3 = \text{the value of } \frac{3n}{4} \text{ observation}$

Measures of Dispersion

Range

The range is simply the difference between the maximum and minimum value in the data.

Example 1 Consider the vector of weight of students in a class.

```
weight <- c(71,69,64,65,73,69,68)

data <- data.frame(weight)

# getting the measures

weight_data <- data|>
  dplyr::summarise(
    min_weight = min(weight),
    mean_weight = mean(weight),
    max_weight = max(weight),
    range = min_weight - max_weight
  )
knitr::kable(weight_data)
```

min_weight	mean_weight	max_weight	range
64	68.42857	73	-9

Variance

The variance is given by the square of the deviations from the mean.

The formula for the variance is given as:

$$Variance(S^2) = \frac{\sum_{i=1}^n (x_i - \bar{x})^2}{n - 1}$$

Example 2 Calculate the variance of the given vector in Example 1.

We can solve this by hand in R as follows:

```
# get the deviations from the mean

weight <- c(71,69,64,65,73,69,68)

# mean and difference

mean_weight <- mean(weight)

# squared differences

diff_mean <- weight-mean_weight

sqd_diff <- diff_mean^2

var_weight <- sum(sqd_diff)/(length(weight)-1)
var_weight
```

```
[1] 9.952381
```

We can use in build R function as follows:

```
weight <- c(71,69,64,65,73,69,68)
var(weight)
```

```
[1] 9.952381
```

Standard Deviation

The standard deviation is the square root of the variance. It returns the variance to its original scale of the variable.

$$S = \sqrt{\frac{\sum_{i=1}^n (x_i - \bar{x})^2}{n - 1}}$$

In the current example

$$S = \sqrt{9.952381} = 3.154$$

In R we use sd function

```
weight <- c(71,69,64,65,73,69,68)
sd(weight)
```

```
[1] 3.154739
```

Kurtosis

Kurtosis is a statistical measure that describes the distribution of data in terms of its tails or outliers compared to a normal distribution.

Kurtosis values can be categorized as follows:

High kurtosis (Leptokurtic)

- A high kurtosis value indicates a distribution with a peak that is more pronounced and sharper than a normal distribution.
- It implies that the data is more concentrated around the mean.
- This can result in a distribution that appears more peaked and less spread out.

Low Kurtosis (Platykurtic)

- A low kurtosis value indicates a distribution with a flatter peak compared to a normal distribution.
- It suggests that the data is more spread out.
- This can result in a distribution that looks flatter and less peaked than a normal distribution.

$$Kurtosis = \frac{n \times \sum_{i=1}^n (Y_i - \bar{Y})^4}{\left(\sum_{i=1}^n (Y_i - \bar{Y})^2 \right)^2}$$

Where:

- Y_i is the i^{th} variable of the distribution
- $\bar{Y}$ is the mean of the distribution
- n number of observations

Example

Use the following data to calculate the kurtosis value and give an interpretation

26	12	16	56	112	24
----	----	----	----	-----	----